JUNIOR LEVEL	PACKET TRACER	LAB 03

Lab 03

Inter-VLAN Routing (Router-on-a-Stick)

Enable communication between VLANs using subinterfaces and 802.1Q encapsulation

Router-on-a-Stick	Subinterfaces	802.1Q Encapsulation	L3 Switch SVIs	TTL Analysis

CCNA Topic	3.5 – Configure and verify Layer 3 routing between VLANs
Exam Objective	Configure router-on-a-stick inter-VLAN routing
Difficulty	Junior
Simulator	Cisco Packet Tracer 8.x
Devices	2911 Router + 2x 2960 Switches + 3560 Multilayer Switch (bonus) + 4x PCs
Prerequisite	Lab 02 — VLAN Configuration & Trunk Links
Date Completed	2026-02-19

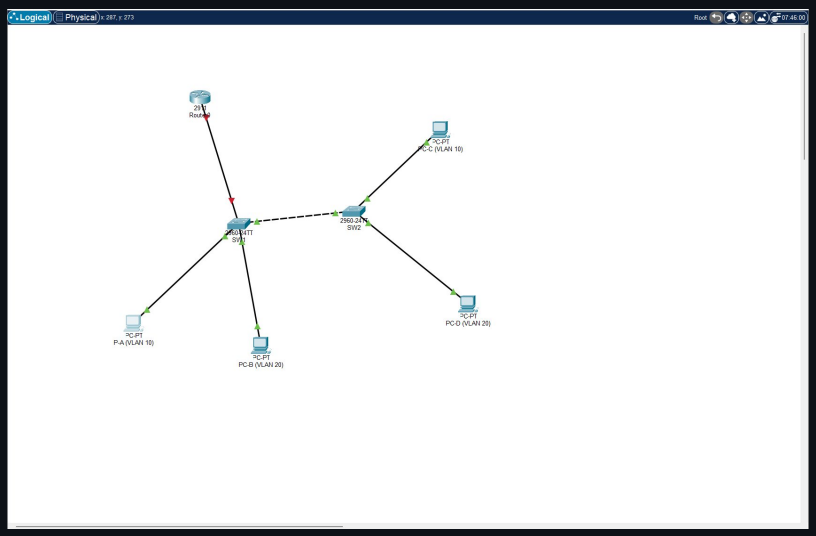

Figure 1 — Lab 03 Packet Tracer topology: R1 router added to Lab 02, linked to SW1 via trunk

Lab Objective

VLANs isolate traffic — but users in different VLANs still need to communicate. Router-on-a-Stick solves this by using a single physical router link with multiple logical subinterfaces, each serving as the default gateway for one VLAN. This lab adds a 2911 router to the Lab 02 topology and configures subinterfaces with 802.1Q encapsulation. The Layer 3 switch SVI alternative was also demonstrated.

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	VLAN
R1	G0/0	unassigned	—	N/A	Trunk
R1	G0/0.10	192.168.10.1	255.255.255.0	N/A	10
R1	G0/0.20	192.168.20.1	255.255.255.0	N/A	20
R1	G0/0.99	192.168.99.1	255.255.255.0	N/A	99
PC-A	Fa0	192.168.10.10	255.255.255.0	192.168.10.1	10
PC-B	Fa0	192.168.20.10	255.255.255.0	192.168.20.1	20
PC-C	Fa0	192.168.10.11	255.255.255.0	192.168.10.1	10
PC-D	Fa0	192.168.20.11	255.255.255.0	192.168.20.1	20

■ The "One Stick" Explained

R1 uses a single physical cable (G0/0) to SW1. Three logical subinterfaces are created on that one physical port — G0/0.10, G0/0.20, and G0/0.99. Each subinterface handles one VLAN. The physical G0/0 has NO IP address — only the subinterfaces have IPs. This single cable is the "stick" in Router-on-a-Stick.

Subinterface Configuration Rules

Configuration Element	Rule / Detail
Physical interface G0/0	Must be no shutdown — do NOT assign an IP address here
Subinterface numbering	Match VLAN ID (G0/0.10 for VLAN 10) — convention, not requirement
encapsulation dot1Q	Mandatory on every subinterface — tells router which VLAN tag to process
Native VLAN keyword	Native VLAN needs: encapsulation dot1Q 99 native
IP address	Each subinterface gets the default gateway IP for its VLAN

Step 1 — SW1: Add Trunk Port for Router (Gi0/2)

SW1 Gi0/2 is configured as a trunk to carry VLANs 10, 20, and 99 to R1. The native VLAN and allowed VLANs must match exactly what is configured on the router subinterfaces.

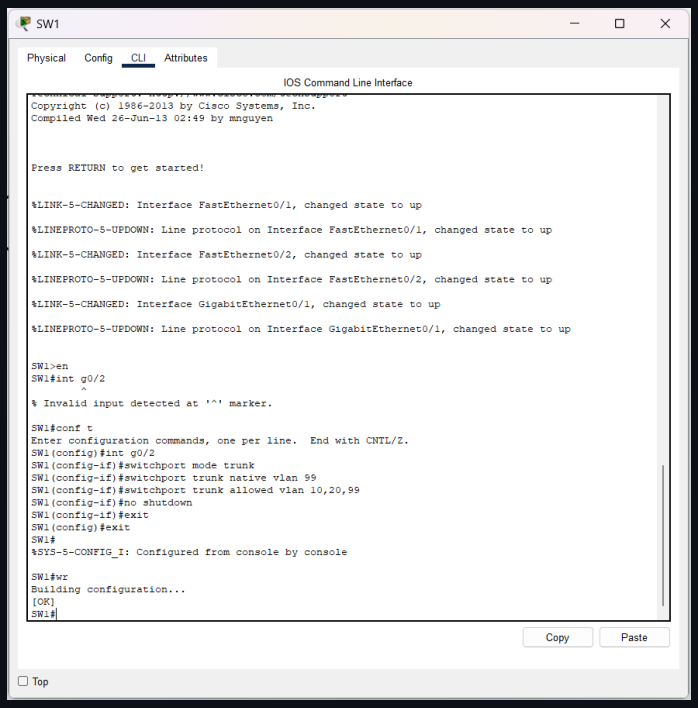

Figure 2 — SW1: Gi0/2 configured as trunk (native 99, allowed 10,20,99), saved with wr. Note: "int g0/2" was first typed at privileged EXEC without conf t — caught and corrected.

■■ Issue 1 — Interface Command at Privileged EXEC

SW1#int g0/2 was typed at privileged EXEC before entering configure terminal. IOS returned: % Invalid input detected at '^' marker. Fix: run conf t first, then int g0/2. All interface configuration commands require global config mode.

SW1 trunk port commands:

```
configure terminal interface GigabitEthernet0/2 switchport mode trunk switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,99 no shutdown exit wr
```

Step 2 — R1: Subinterface Configuration

Three subinterfaces are created on G0/0 — one per VLAN. The physical G0/0 is brought up with no shutdown but carries no IP. Each subinterface needs an encapsulation statement and an IP address.

Figure 3 — R1: hostname set, G0/0 up with no IP, all three subinterfaces configured with dot1Q encapsulation and IPs. Config saved with wr.

■■ Issue 2 — wr Typed Inside Global Config Mode

R1(config)#wr was entered while still inside configure terminal. IOS returned: % Invalid input detected. Fix: use end to return to R1# privileged EXEC first, then run wr. The copy run start / wr command only works from privileged EXEC.

R1 subinterface configuration:

```
enable configure terminal hostname R1 interface GigabitEthernet0/0 no shutdown exit interface
GigabitEthernet0/0.10 encapsulation dot1Q 10 ip address 192.168.10.1 255.255.255.0 exit interface
GigabitEthernet0/0.20 encapsulation dot1Q 20 ip address 192.168.20.1 255.255.255.0 exit interface
GigabitEthernet0/0.99 encapsulation dot1Q 99 native ip address 192.168.99.1 255.255.255.0 exit end wr
```

Step 3 — PC Default Gateway Update

All four PCs are updated with default gateways pointing to their VLAN's router subinterface. This is the critical change from Lab 02 — without a gateway, cross-VLAN packets have no path to the router.

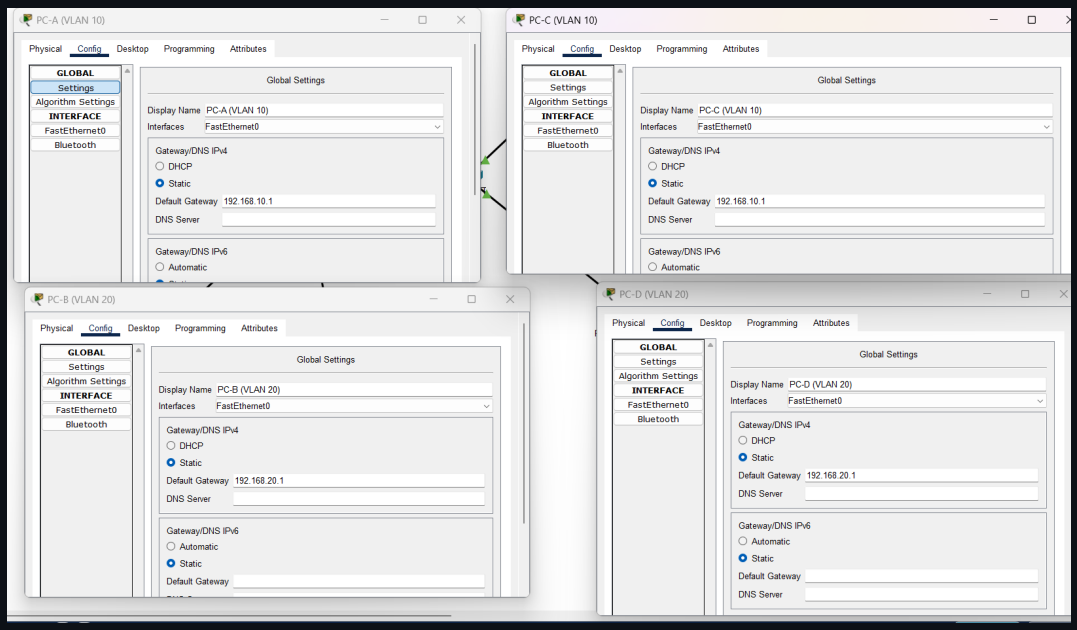

Figure 4 — All four PCs with default gateways configured: PC-A and PC-C → 192.168.10.1 (VLAN 10), PC-B and PC-D → 192.168.20.1 (VLAN 20)

PC Gateway Configuration Table

PC	IP Address	Subnet Mask	Default Gateway	VLAN
PC-A	192.168.10.10	255.255.255.0	192.168.10.1	10 — SALES
PC-B	192.168.20.10	255.255.255.0	192.168.20.1	20 — ENGINEERING
PC-C	192.168.10.11	255.255.255.0	192.168.10.1	10 — SALES
PC-D	192.168.20.11	255.255.255.0	192.168.20.1	20 — ENGINEERING

Step 4 — Inter-VLAN Routing Verification

Three pings and a tracert were run from PC-A. The TTL value is the key diagnostic: same-VLAN pings return TTL=128 (switched at Layer 2, no router hop), while cross-VLAN pings return TTL=127 (one router hop reduces TTL by 1). The tracert confirms R1 is the gateway for cross-VLAN traffic.

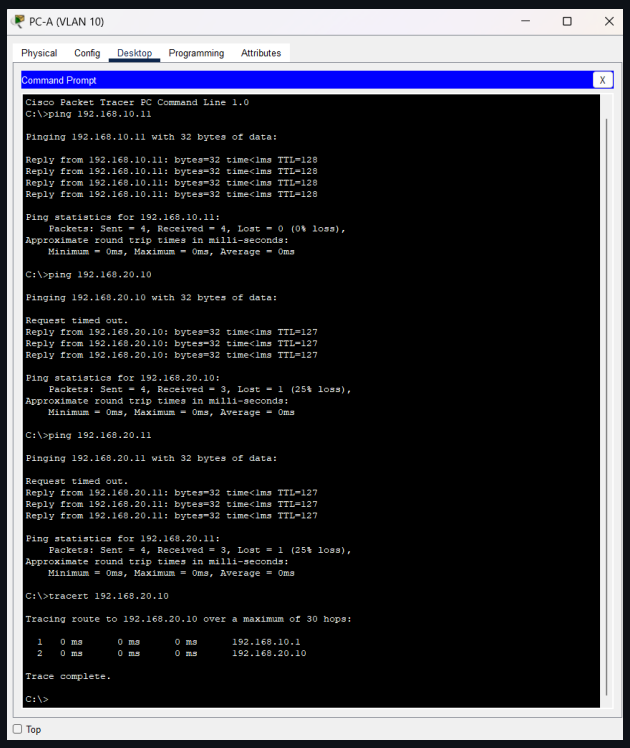

Figure 5 — PC-A: ping PC-C same VLAN (TTL=128), ping PC-B cross VLAN (TTL=127), ping PC-D cross VLAN (TTL=127), tracert confirms R1 as hop 1

Results Summary

Test	Destination IP	Path	Ping Result	TTL / Notes
PC-A → PC-C	192.168.10.11	Same VLAN 10	4/4 — 0% loss	128 — switched (no router)
PC-A → PC-B	192.168.20.10	Cross VLAN 10→20	3/4 — ARP delay	127 — routed via R1
PC-A → PC-D	192.168.20.11	Cross VLAN 10→20	3/4 — ARP delay	127 — routed via R1
tracert PC-B	192.168.20.10	Cross VLAN	Hop1: 192.168.10.1 Hop2: 192.168.20.10	R1 confirmed

■ TTL Diagnostic — How to Confirm Routing Occurred

TTL=128: packet was switched at Layer 2 — same VLAN, no router involved. TTL=127: packet went through one router hop — inter-VLAN routing via R1. Each router hop decrements TTL by 1. This is a reliable exam and real-world confirmation that Router-on-a-Stick is working correctly. The 25% loss (1/4 packets) on cross-VLAN pings is normal — the first packet triggers ARP resolution on the new subnet.

Step 5 — R1 Verification Show Commands

Three show commands confirm the complete subinterface configuration: show ip interface brief shows all subinterfaces up/up with correct IPs, show ip route confirms all three subnets are reachable, and show interfaces G0/0.10 confirms the 802.1Q encapsulation is active.

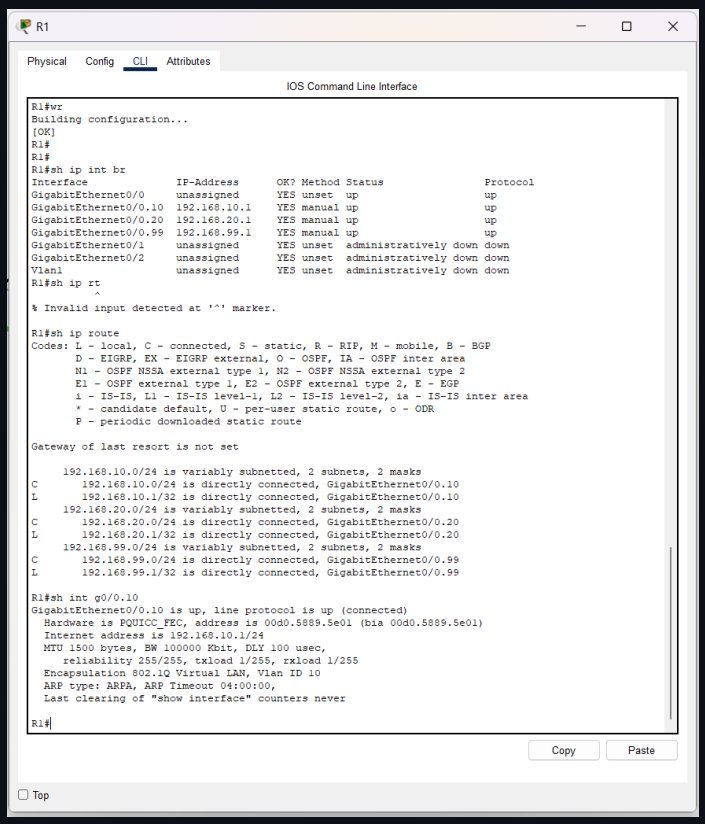

Figure 6 — R1: show ip int brief (all subinterfaces up/up), show ip route (3 connected subnets), show int g0/0.10 (802.1Q VLAN ID 10 confirmed). Note: "sh ip rt" abbreviation was rejected — corrected to "show ip route".

■■ Issue 3 — "sh ip rt" Abbreviation Not Recognised

"sh ip rt" was typed as shorthand for show ip route. IOS returned: % Invalid input detected at '^' marker. Fix: use the full command show ip route, or the valid abbreviation sh ip ro. IOS abbreviations only work when the partial string is unambiguous.

Show Command Reference

Command	Expected Output	What It Confirms
show ip interface brief	G0/0 unassigned up/up All subinterfaces up/up with correct IPs	All subinterfaces active and correctly assigned
show ip route	C 192.168.10.0/24 via G0/0.10 C 192.168.20.0/24 via G0/0.20 C 192.168.99.0/24 via G0/0.99	All 3 VLAN subnets in routing table
show interfaces G0/0.10	Encapsulation 802.1Q Virtual LAN, Vlan ID 10 Internet address 192.168.10.1/24	802.1Q active with correct VLAN ID

Bonus — Layer 3 Switch SVI Alternative

Going beyond the lab requirements, the SVI routing method was configured on a Cisco 3560-24PS Multilayer Switch. This is the preferred method for production networks — routing happens in dedicated hardware ASICs at full wire speed, with no single-link bottleneck.

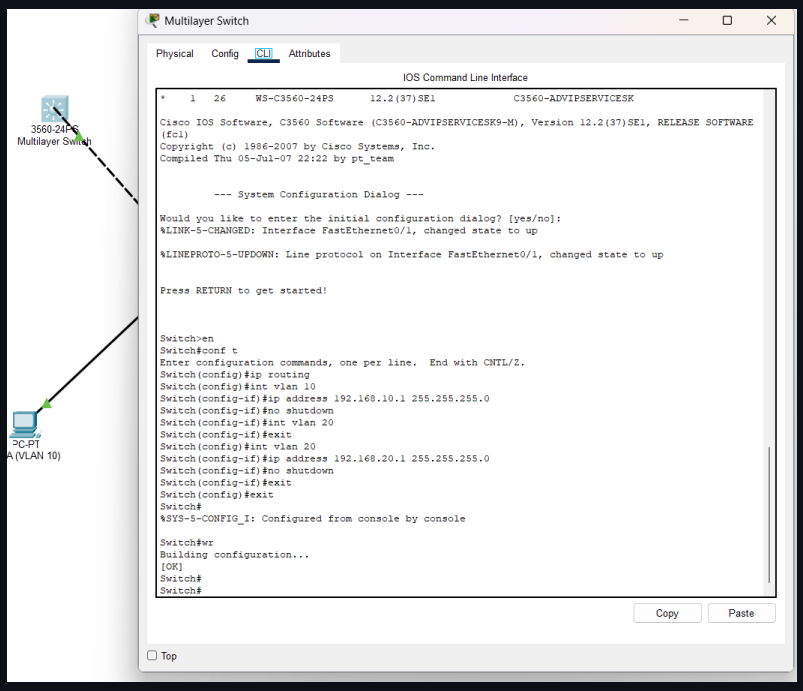

Figure 7 — Cisco 3560-24PS: ip routing enabled, VLAN 10 and VLAN 20 SVIs configured with gateway IPs 192.168.10.1 and 192.168.20.1, saved. Demonstrates the L3 switch alternative.

Layer 3 Switch SVI Configuration

```
enable configure terminal ip routing interface vlan 10 ip address 192.168.10.1 255.255.255.0 no shutdown exit
interface vlan 20 ip address 192.168.20.1 255.255.255.0 no shutdown exit wr
```

ROAS vs Layer 3 Switch

Feature	Router-on-a-Stick	Layer 3 Switch (SVI)
Hardware	External 2911 router	3560 multilayer switch
Routing speed	Software / CPU	Hardware ASIC — faster
Bandwidth	Single uplink bottleneck	Full switch backplane
Configuration	Subinterfaces + dot1Q	ip routing + interface vlan
Best use case	Small networks / lab / exam	Enterprise production
CCNA exam	■ Tested	■ Tested

Key Takeaways

Physical Interface Has No IP

G0/0 must be no shutdown but carries no IP address. All IPs live on the subinterfaces. Assigning an IP to the physical interface is the most common ROAS mistake.

encapsulation dot1Q is Mandatory

Every subinterface needs encapsulation dot1Q [vlan-id] before the ip address command. Without it, the router does not associate the subinterface with any VLAN.

Native VLAN Needs the "native" Keyword

encapsulation dot1Q 99 native — the native keyword is required on the native VLAN subinterface. Omitting it causes a mismatch with the switch trunk.

TTL=127 Proves Routing Happened

Cross-VLAN pings return TTL=127 (one less than the default 128). Same-VLAN pings stay at TTL=128. This single decrement is reliable proof that ROAS is working.

wr Must Be Run from Privileged EXEC

copy run start and wr only work from the R1# prompt. Typing them inside configure terminal produces an invalid input error.

Switch Port to Router Must Be a Trunk

SW1 Gi0/2 must be a trunk with matching native VLAN and allowed VLANs. An access port will only pass one VLAN — ROAS needs all VLANs on that link.

Layer 3 Switches Are Better for Production

ROAS creates a bandwidth bottleneck on one physical link. Layer 3 switches route in hardware at full wire speed — no bottleneck, no external router needed.

Lab Validation Checklist

- ✓ R1 added to topology — cabled to SW1 Gi0/2
- ✓ SW1 Gi0/2: trunk mode, native VLAN 99, allowed 10,20,99
- ✓ R1 G0/0: no shutdown — no IP address on physical interface
- ✓ R1 G0/0.10: encapsulation dot1Q 10, IP 192.168.10.1/24
- ✓ R1 G0/0.20: encapsulation dot1Q 20, IP 192.168.20.1/24
- ✓ R1 G0/0.99: encapsulation dot1Q 99 native, IP 192.168.99.1/24
- ✓ PC-A and PC-C gateways updated to 192.168.10.1
- ✓ PC-B and PC-D gateways updated to 192.168.20.1
- ✓ PC-A → PC-C (same VLAN 10): 0% loss, TTL=128 ■
- ✓ PC-A → PC-B (cross VLAN): success, TTL=127 confirms routing ■
- ✓ PC-A → PC-D (cross VLAN): success, TTL=127 confirms routing ■
- ✓ tracert confirms R1 (192.168.10.1) as hop 1 for cross-VLAN traffic ■
- ✓ show ip route: all 3 subnets present via subinterfaces ■
- ✓ show interfaces G0/0.10: 802.1Q VLAN ID 10 confirmed ■
- ✓ Layer 3 switch SVI method demonstrated on Cisco 3560 (bonus) ■

Lab 03 Complete — Inter-VLAN routing is tested on every CCNA exam.

Next: Lab 04 — Static Routing